

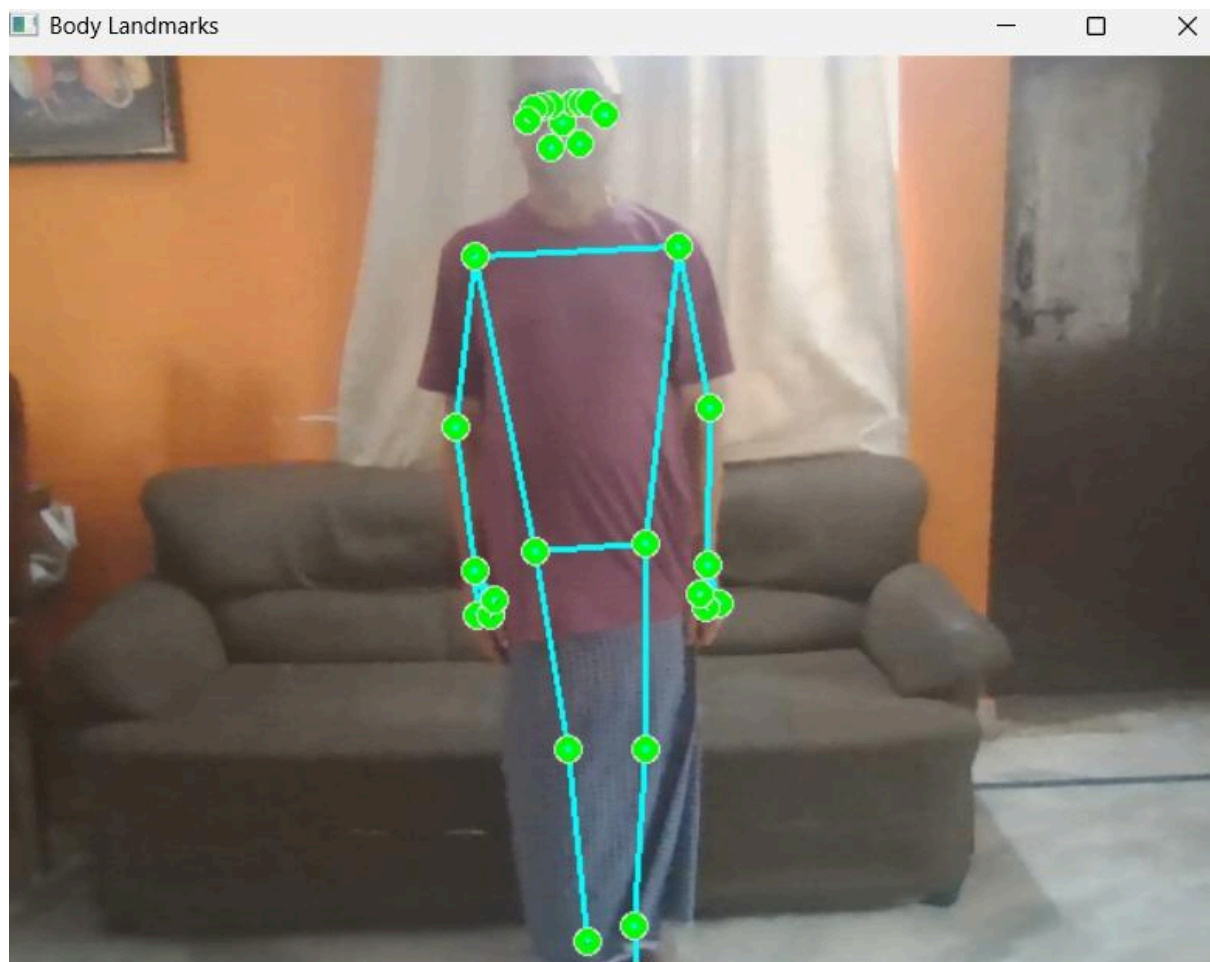
Navya Reddy

AI ML

Day 7 task

1.Landmark Detection

The output of landmark detection is a real-time webcam display where the system detects and tracks important body points such as the nose, shoulders, elbows, wrists, hips, knees, and ankles using MediaPipe and OpenCV. Small circles are drawn on each detected landmark, and lines connect them to form a human skeleton structure on the screen. The program can also print the landmark IDs and their x-y coordinates in the console, helping track body movement and posture accurately. This output is commonly used in applications like fitness tracking, gesture recognition, pose estimation, and motion analysis.



2.pose detection

```
import cv2
import mediapipe as mp
import os
```

```

image_path = r"C:\Users\navya\Downloads\person.png"
# Replace with YOUR image path
# Example:
# image_path = r"C:\Users\navya\Downloads\Screenshot.png"

# ----- CHECK -----
print("Path exists:", os.path.exists(image_path))

if not os.path.exists(image_path):
    print("Image not found wrong")
    exit()

# ----- LOAD -----
img = cv2.imread(image_path)

if img is None:
    print("OpenCV cannot read image wrong")
    exit()

h, w, _ = img.shape

# ----- POSE -----
mp_pose = mp.solutions.pose
mp_draw = mp.solutions.drawing_utils

with mp_pose.Pose(
    static_image_mode=True,
    min_detection_confidence=0.5
) as pose:

    rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

    results = pose.process(rgb)

    if results.pose_landmarks:

        mp_draw.draw_landmarks(
            img,
            results.pose_landmarks,
            mp_pose.POSE_CONNECTIONS
        )

        lm = results.pose_landmarks.landmark

```

```

def px(i):
    return (
        int(lm[i].x * w),
        int(lm[i].y * h)
    )

shoulder = abs(px(12)[0] - px(11)[0])
waist = abs(px(24)[0] - px(23)[0])
height = abs(px(27)[1] - px(11)[1])

cv2.putText(
    img,
    f"Height: {height}px",
    (20, 40),
    cv2.FONT_HERSHEY_SIMPLEX,
    0.8,
    (0,255,0),
    2
)

cv2.putText(
    img,
    f"Shoulder: {shoulder}px",
    (20, 80),
    cv2.FONT_HERSHEY_SIMPLEX,
    0.8,
    (0,255,0),
    2
)

cv2.putText(
    img,
    f"Waist: {waist}px",
    (20,120),
    cv2.FONT_HERSHEY_SIMPLEX,
    0.8,
    (0,255,0),
    2
)

print("Pose detected correct")

```

else:

```
print("No body detected wrong")
```

```
# ----- SHOW -----
```

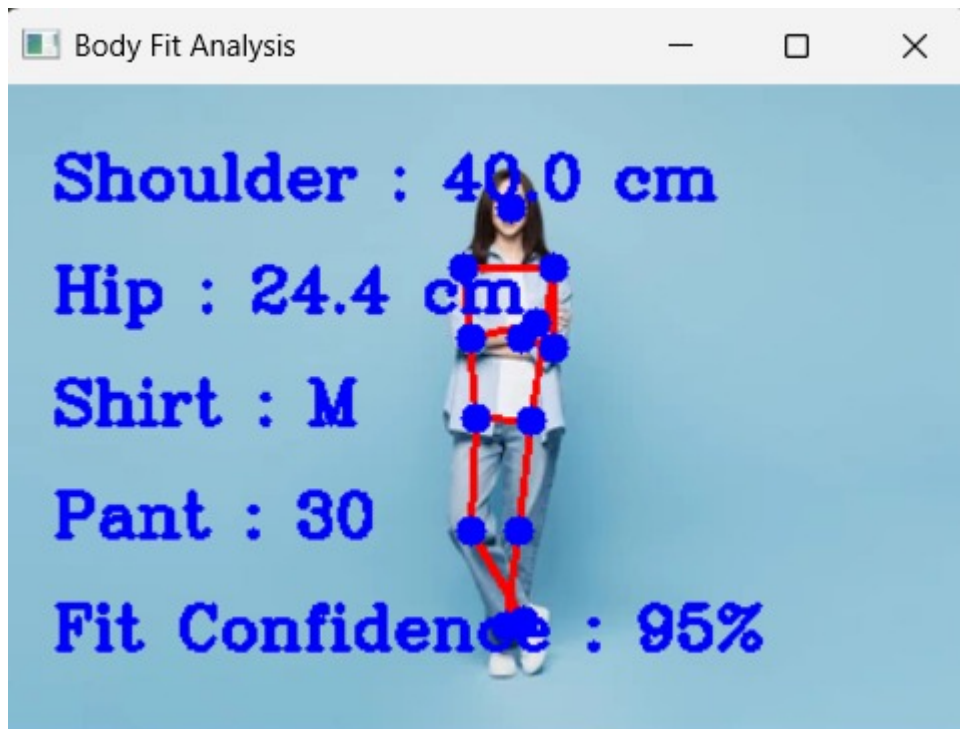
```
cv2.imshow("Human Pose Detection", img)
```

```
cv2.waitKey(0)
```

```
cv2.destroyAllWindows()
```

Out put

The image is generated by a pose measurement and recommendation system that detects body posture and key body points using computer vision techniques. The system measures different body parts such as shoulders, arms, legs, torso, and neck by connecting detected joints with lines and points on the body. Based on these measurements, it provides clothing recommendations like suitable shirt and pant categories. This type of application is useful for virtual fitting systems, online shopping assistance, posture analysis, and smart retail solution



3.Multi angle pose

```
import cv2
import mediapipe as mp
import math
mp_pose = mp.solutions.pose
pose = mp_pose.Pose(
    static_image_mode=True,
    min_detection_confidence=0.5
)
```

```

mp_draw = mp.solutions.drawing_utils
image_path = r"C:\Users\91630\Downloads\WhatsApp Image 2026-05-19 at
3.32.15 PM.jpeg"
image = cv2.imread(image_path)
if image is None:
    print("Image not found")
    exit()
original = image.copy()
rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)
results = pose.process(rgb)
h, w, _ = image.shape
def distance(p1, p2):
    return math.sqrt(
        (p2[0] - p1[0])**2 +
        (p2[1] - p1[1])**2
    )
if results.pose_landmarks:
    landmarks = results.pose_landmarks.landmark
    left_shoulder = (
        int(landmarks[11].x * w),
        int(landmarks[11].y * h)
    )
    right_shoulder = (
        int(landmarks[12].x * w),
        int(landmarks[12].y * h)
    )
    left_hip = (
        int(landmarks[23].x * w),
        int(landmarks[23].y * h)
    )
    right_hip = (
        int(landmarks[24].x * w),
        int(landmarks[24].y * h)
    )
    left_ankle = (
        int(landmarks[27].x * w),
        int(landmarks[27].y * h)
    )
    right_ankle = (

```

```

        int(landmarks[28].x * w),
        int(landmarks[28].y * h)
    )
    shoulder_width = distance(
        left_shoulder,
        right_shoulder
    )

    hip_width = distance(
        left_hip,
        right_hip
    )

    left_leg = distance(
        left_hip,
        left_ankle
    )

    right_leg = distance(
        right_hip,
        right_ankle
    )
    mp_draw.draw_landmarks(
        original,
        results.pose_landmarks,
        mp_pose.POSE_CONNECTIONS,
        mp_draw.DrawingSpec(
            color=(0,255,0),
            thickness=3,
            circle_radius=3
        ),
        mp_draw.DrawingSpec(
            color=(255,0,0),
            thickness=2
        )
    )
    cv2.rectangle(
        original,
        (20,20),
        (420,190),
        (0,0,0),
        -1
    )

```

```
cv2.putText(
    original,
    "MULTI-ANGLE HUMAN POSE ESTIMATION",
    (30,50),
    cv2.FONT_HERSHEY_SIMPLEX,
    0.7,
    (0,255,255),
    2
)

cv2.putText(
    original,
    f"Shoulder Width : {shoulder_width:.2f}",
    (30,90),
    cv2.FONT_HERSHEY_SIMPLEX,
    0.7,
    (255,255,255),
    2
)

cv2.putText(
    original,
    f"Hip Width      : {hip_width:.2f}",
    (30,120),
    cv2.FONT_HERSHEY_SIMPLEX,
    0.7,
    (255,255,255),
    2
)

cv2.putText(
    original,
    f"Left Leg Length : {left_leg:.2f}",
    (30,150),
    cv2.FONT_HERSHEY_SIMPLEX,
    0.7,
    (255,255,255),
    2
)

cv2.putText(
    original,
```

```

        f"Right Leg Length: {right_leg:.2f}",
        (30,180),
        cv2.FONT_HERSHEY_SIMPLEX,
        0.7,
        (255,255,255),
        2
    )

    print("Pose detected successfully")

else:
    print("No pose detected")

output_path = r"C:\Users\91630\Downloads\full_pose_output.jpg"

cv2.imwrite(output_path, original)

print("Saved at:")
print(output_path)
cv2.namedWindow(
    "Pose Estimation",
    cv2.WINDOW_NORMAL
)

cv2.imshow(
    "Pose Estimation",
    original
)

cv2.waitKey(0)
cv2.destroyAllWindows()

```

Out put

MULTI-ANGLE HUMAN POSE ESTIMATION

Shoulder Width : 182.21
Hip Width : 114.07
Left Leg Length : 495.50
Right Leg Length: 472.02

